Year 10 Mathematics

Algebra 2

7.3 Solving Equations with Roots and Powers

Solving Equations with Roots and Powers

STARTER · 5 MIN

Date: _____ **Title:** Solving Equations with Roots and Powers

LEARNING OBJECTIVES

- Solve equations involving squares and cubes
- Solve equations involving square roots
- Solve harder equations with powers and roots

STARTER *Answer in your book. No calculators.*

1. Work out $\sqrt{64}$

2. Work out $\sqrt[3]{27}$

3. Work out 5^2

4. Solve $3x - 4 = 11$

Starter — Answers

STARTER

Tick or correct in green pen

1. $\sqrt{64} = \mathbf{8}$

2. $\sqrt[3]{27} = \mathbf{3}$

3. $5^2 = \mathbf{25}$

4. $3x = 15, \quad x = \mathbf{5}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can solve equations involving squares and cubes	☐
SILVER	I can solve equations involving square roots	☐
GOLD	I can solve harder equations with powers and roots	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Roots and Powers

INTRODUCTION · 3 MIN

Undo a power with a root, and undo a root with a power.

Squares

Square root both sides. There are two answers.

$x^2 = 36, \quad x = \pm 6$

Cubes

Cube root both sides. There is one answer.

$x^3 = 125, \quad x = 5$

Square roots

Square both sides.

$\sqrt{x} = 7, \quad x = 49$

Order

Get the power or root on its own first.

$2x^2 = 50 \rightarrow x^2 = 25$

Examples 1 and 2 — Squares and Cubes

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Solve

$x^2 = 49$

Example 2

Solve

$x^3 = 64$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$x = \pm\sqrt{49}$

$x = \pm 7$

Tip: x can be positive or negative.

Example 2

$x = \sqrt[3]{64}$

$x = 4$

Tip: a cube has only one real root.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve each equation*

1. $x^2 = 81$
2. $x^2 = 144$
3. $x^3 = 8$
4. $x^3 = -27$
5. $y^2 = 0.25$

★ Solve $\sqrt{x} = 6$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $x = \pm 9$

2. $x = \pm 12$

3. $x = \sqrt[3]{8} = 2$

4. $x = \sqrt[3]{-27} = -3$

5. $y = \pm 0.5$

★ $x = 6^2 = 36$

Examples 3 and 4 — Rearranging First

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Solve

$3x^2 = 75$

Example 4

Solve

$\sqrt{x} + 4 = 9$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x^2 = 25$$

$$x = \pm 5$$

Tip: divide by 3 first.

Example 4

$$\sqrt{x} = 5$$

$$x = 25$$

Tip: get the root on its own, then square.

Task 2

PRACTICE · 9 MIN

SILVER

Solve each equation

1. $2x^2 = 32$

2. $x^2 + 5 = 41$

3. $5x^3 = 40$

4. $\sqrt{x} - 3 = 4$

5. $3\sqrt{x} = 12$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $x^2 = 16, \quad x = \mathbf{\pm 4}$

2. $x^2 = 36, \quad x = \mathbf{\pm 6}$

3. $x^3 = 8, \quad x = \mathbf{2}$

4. $\sqrt{x} = 7, \quad x = \mathbf{49}$

5. $\sqrt{x} = 4, \quad x = \mathbf{16}$

Examples 5 and 6 — Harder Equations

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (3 marks)

Solve

$(x+3)^2 = 25$

Example 6 (3 marks)

Solve

$\sqrt{2x+1} = 5$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$x + 3 = \pm 5$$

$$x = \mathbf{2} \text{ or } \mathbf{-8}$$

Tip: square root both sides, then solve two equations.

Example 6

$$2x + 1 = 25$$

$$2x = 24$$

$$x = \mathbf{12}$$

Tip: square both sides first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Solve $(x-2)^2 = 49$ (3 marks)

2. Solve $\sqrt{3x-2} = 4$ (3 marks)

3. Solve $2(x+1)^3 = 54$ (3 marks)

4. Solve $\frac{x^2}{4} = 9$ (2 marks)

5. A square has an area of 60 cm². Find the length of one side to 3 s.f. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $x - 2 = \pm 7, \quad x = \mathbf{9}$ or $\mathbf{-5}$

2. $3x - 2 = 16, \quad 3x = 18, \quad x = \mathbf{6}$

3. $(x + 1)^3 = 27, \quad x + 1 = 3, \quad x = \mathbf{2}$

4. $x^2 = 36, \quad x = \mathbf{\pm 6}$

5. $x^2 = 60, \quad x = \sqrt{60} = \mathbf{7.75}$ cm

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $x^2 = 100$

SILVER Solve $4x^2 = 36$

GOLD Solve $\sqrt{x+5} = 3$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x = \pm 10$

SILVER $x^2 = 9, \quad x = \pm 3$

GOLD $x + 5 = 9, \quad x = 4$

Exam Practice — Squares and Cubes

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

Solve

(a) (1)

$x^2 = 49$

x = ...

(b) (1)

$x^3 = 64$

x = ...

Question 2

(2 marks)

Solve

$2x^2 = 72$

x = ...

Exam Practice — Rearranging First

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(3 marks)

Solve

$$3x^2 - 5 = 70$$

x = ...

Question 4

(4 marks)

Solve

(a) (1)

$$\sqrt{x} = 6$$

x = ...

(b) (3)

$$\sqrt{2x - 1} = 3$$

x = ...

Exam Practice — Harder Equations

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Solve

$$(x - 2)^2 = 81$$

.....

Question 6

(3 marks)

Solve

$$\frac{3x^2}{2} - 4 = 20$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) x = 7 or x = −7 — B1

(b) x = 4 — B1

Tip: a square has TWO roots; a cube has one.

Question 2

$x^2 = 36$ — M1

x = ±6 — A1

Tip: divide by 2 before square-rooting.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$3x^2 = 75$ — M1

$x^2 = 25$ — M1

$x = \pm 5$ — A1

Tip: undo in reverse: +5, ÷3, then square root.

Question 4

(a) $x = 36$ — B1

(b) $2x - 1 = 9$ — M1

$2x = 10$ — M1

$x = 5$ — A1

Tip: square both sides to remove the root.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x - 2 = \pm 9$ — M1

$x = 2 + 9$ or $x = 2 - 9$ — M1

$x = 11$ or $x = -7$ — A1

Tip: square root both sides first.

Question 6

$\frac{3x^2}{2} = 24$ — M1

$3x^2 = 48$, $x^2 = 16$ — M1

$x = \pm 4$ — A1

Tip: +4, ×2, ÷3, then square root.